PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT AUTHORITY
KINONDONI MUNICIPAL COUNCIL
FORM FOUR DISTRICT PRE-MOCK EXAMINATION

CODE: 032 **CHEMISTRY 1**

Time: 3 Hours **Year: March 2024**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and any **TWO (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Non-programmable calculators may be used.
5. Remember to write your number on each page of your answer sheet.
6. The following constants may be used:

Atomic masses: S = 32, O = 16, N = 14, H = 1, C = 12, Cl = 35.5
$1dm^3 = 1000cm^3 = 1$ Litre
GMV at STP = $22.4\ dm^3$

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SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter beside the item number in the answer space provided.

(i) An industrial preparation of ethane from higher alkane is done by;-

A. Sublimation C. Distillation D. Synthesis
B. Cracking E. Ethenelyzation

(ii) In ammonia gas, hydrogen and nitrogen atoms are held together by:-

A. Covalent bond D. Electrovalent bond
B. Weak force of attraction E. Ionization bond
C. Ionic bond

(iii) The volume occupied by 8.4g of nitrogen gas at S.T.P is;-

A. 22.4 dm^3 B. 22400 dm^3 C. 13.44 dm^3 D. 6720 dm^3 E. 67.20 dm^3

(iv) When an atom loose electron it becomes;-

A. Molecule B. Isotope C. Cation D. Proton E. Anion

(v) A quantity of electricity is measured in:-

A. Ampere B. Watts C. Coulombs D. Seconds E. Joules

(vi) Any substance produced and released into the environment that is; harmful is called;-

A. Waste C. Pollution D. Polluter
B. Pollutants E. Sewage

(vii) Existence of two or more atoms with the same atomic number but different mass number is called:-

A. Isotopy C. Allotropes D. Isobars
B. Isotopes E. Allotropy

(viii) How can you obtain iron filling from the mixture of solid iodine and iron filling?

A. Evaporation C. Chromatography D. Decantation
B. Fractional distillation E. Sublimation

(ix) The primary macro elements needed by plants for proper growth and development are:-

A. N, S and Ca B. P, N and C C. P, N and K D. N, P and Ca E. N, Mg and Mn

(x) A piece of land has a soil with PH of 6.2, this indicates that the soil is:-

A. High acidic C. Both acid and neutral
B. Strong basic D. Slightly acidic E. Slightly basic

2. Match the items in **List A** with those in **List B** then write the correct letter on the space provided.

LIST A	**LIST B**
(i) The extinguisher composed of bromochloro-diflouro methane	A. Dry chemical
(ii) The extinguisher composed of potassium acetate	B. Fire blanket extinguisher
(iii) The extinguisher composed of fine sodium bicarbonate powder pressurized by nitrogen	C. APW extinguisher
(iv) The extinguisher composed of mono-ammonium phosphate with a nitrogen carrier	D. Carbon dioxide extinguisher
(v) The extinguisher composed of tap water pressurized by air	E. Halon extinguisher
	F. ABC extinguisher
	G. Sand bucket extinguisher
	H. Wet chemical extinguisher
	I. Foam extinguisher

Vanadium pentoxide

SECTION B (54 marks)
Answer all questions

3. (a) Abdul wanted to test a substance Q using blue cobalt (II) chloride. When cobalt (II) chloride was exposed to the substance, it changed from blue to pink. Use this information to answer the following questions:-
   (i) Name the substance that was tested.
   (ii) Name other substance that could be used in place of cobalt (II) chloride.
   (iii) Briefly explain three ways by which the substance can be purified.

   (b) Describe the energy transformation that takes place;
   (i) Energy from the sun used to generate electricity for lighting a house.
   (ii) Mechanical energy from the waterfall is used to generate electricity.

4. (a) State the Le chatelliers principle.
   (b) In the industrial preparation of Sulphur trioxide equilibrium is established between Sulphur dioxide and oxygen gas

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} \quad \Delta H = -94.9^{KJ/mol}$$

   (i) How would you adjust temperature and pressure to maximize the proportion of product at equilibrium?
   (ii) Why is it unfavourable to work with very high pressure and very low temperature in contact process.
   (iii) What catalyst is used to speed up the rate of formation of Sulphur trioxide before attaining equilibrium.

5. (a) How many moles of nitrogen gas would react with excess hydrogen gas to produce 0.56 dm^3 of ammonia gas at S.T.P.
   (b) Ackim Ziech took egg shells and put them inside the beaker containing hydrochloric acid. Abruptly the reaction started to take place inside the beaker.
   (i) Name the reaction taking place inside the beaker.
   (ii) Balance the ionic equation of the reaction taking place.
   (iii) Outline any three application of the process named in b(i) above in the blast furnace.

6. Extraction of Iron from its Ore passes through different zones such as lower, zone, middle zone and upper zone.
   (i) Write three equations that took place in the middle zone.
   (ii) Explain the purpose of adding limestone and coke in the blast furnance.

7. (a) A metallic object coated with copper is placed in $CuSO_4$ electrolyte.
   (i) To which electrode should the object be connected where direct current power supply flow?
   (ii) What current would be required to produce 1120cm^3 of oxygen gas by the electrolysis at S.T.P for one hour and 30 minutes?

   (b) What would happen if the principles of electrolysis were not discovered up to now? Give two reasons.

8. (a) Explain the following.
   (i) Why group I elements are called alkali metals.
   (ii) Why non-metals are called oxidizing agents.

   (b) The reaction between dilute hydrochloric acid and sodium thiosulphate results in the production of white precipitates which cause the opaqueness of the solution. Mention any three uses of solid substance which cause the opaqueness of the solution.

SECTION C (30 marks)

Answer **Two (02)** questions from this section.

9. (a) When a diluted nitric acid is added to green solid P, a blue solution X is formed with evolution a gas R that forms a white precipitate with limewater. Heating the blue solution of the hydrated salt X, it initially loses water then decomposes to form black solid S, Brown fumes of gas Y and gas Z.
   (i) Identify the solid X, gas Y and gas Z.
   (ii) Write an equation for the reaction between solid P and dilute nitric acid.
   (iii) Write an equation for formation of solid S.

   (b) What will be observed when;
   (i) Manganese (IV) oxide react with concentrated hydrochloric acid.
   (ii) Chlorine gas reacts with hot potassium hydroxide solution.
   (iii) Sulphur reacts with hot concentrated nitric acid.
   (iv) Hydrogen sulphide reacts with sulphur dioxide.
   (v) Concentrated sulphuric acid added to sugar.

10. (a) (i) What is Substitution reaction.
   (ii) By indicating the necessary conditions show steps for complete chlorination of methane.

   (b) Mwaisera Mtumbali a form four student at Busale Secondary School made a mistake to give the correct IUPAC names of organic compound given by his teacher as shown below;

   (i) $CH_2\,CH_2CH_2CH_2\,CH_3$ (with CH_3 branch) name: 1 – methyl pentane

   (ii) $CH_3\,CH_2CH = CHCH_3$ name: Pent -3-ene

   (iii) $CH_3\,CH_2\,CHC = CCH_3$ (with Br) Name: 3-bromohex -4-yne

   (iv) $CH_3\,CH_2\,C\;C - CH_2 - OH$ (with Br, Cl) Name: 2-chloro-3bromo pent-1-ol

   (v) $CH_3\,CH\,CH_2\,COOH$ (with CH_2, CH_3) Name: 3-ethy butanoic acid

   As an expert in organic chemistry show how you can help him by identifying the mistake and give the correct IUPAC name on each compound.

11. (a) (i) Arrange the alkaline earth metals available in the periodic table of the first twenty elements as they appear from the top to the bottom. Explain briefly why the atomic radius increases as one is moving down the alkaline earth metals.
   (iii) Explain the concept of reducing power of metal and state its trend across the period and down the group.

   (b) You are provided with element R and S whose atomic numbers are 20 and 17 respectively.
   (i) Write the formula of the compound and predict the nature of the bond formed when R combines with S.
   (ii) State three properties of the compound formed when R combines with S.